Algorithms Notes

#1 - $T(1)=8$ start at n & work down

- $T(n) = 3T(n-1) - 15$
- $3(3T(n-2)-15)-15$ - $3^2(3T(n-3)-15) - 3\cdot 15 - 15$

$T(n-1) = 3T(n-2) - 15$ - $3^4(3T(n-4) - 3^3 \cdot 15 - 3^2 \cdot 15 - 3\cdot 15 - 15$

$T(n-2) = 3T(n-3) - 15)$

$3^{n-1}\ n-(n-1) - 3^{n-2} \cdot 15 - 3^{n-3} \cdot 15$

$3^{n-1} T(1) - 15 \sum_{i=0}^{n-2} 3^i$

$3^{n-1} T(1) - 15\left(\frac{3^{n-1}-1}{2}\right)$

$$\frac{3^{n-1}+15}{2}$$

- - Master's Method
  - $T(n) = aT(n/b) + f(n)$
  - a & b must be constant - $a \geq 1,\ b > 1$
  - $f(n) = O(n^{\log_b(a) - \varepsilon})$, $\varepsilon > 0$
  - three cases...
  - Example 1 - $T(n) = T(n/2) + 3$

    $a=1,\ b=2,\ f(n)=3$

    $\lg 1 = 0$

    $3 = O(n^{0-\varepsilon})$, $\varepsilon > 0$

    $3 = O(1)$ - case 1 doesn't work

$1 = \Theta(n^0)$, falls into case 2
$T(n) = \Theta(n^0 \lg n) = \Theta(\lg n)$

- Example 2
  $a = 9$, $b = 3$, $f(n) = n$
  $\log_b a = \log_3 9 = 2$
  $n = O(n^{2-\varepsilon})$, $\varepsilon > 0$
  case 1 is valid so $T(n) = \Theta(n^{\log_b a})$

- $T(n) = 2T(n/2) + n \lg n$
  $b = 2$, $a = 2$, $\log_b a = 1$
  $f(n) = n \lg n$
  case 1 $n \lg n = O(n^{1-\varepsilon})$, $\varepsilon > 0$ false
  case 2 $n \lg n = \Theta(n^1)$, $\varepsilon > 0$ false
  case 3 $n \lg n = \Omega(n^{1+\varepsilon})$, $\varepsilon > 0$

  $n < n \lg n < n^{1+\varepsilon}$
  $\lg n$ is bigger than a constant, but smaller than a power

- Multiplying Integers
- usually takes n^2 to multiply
- $A = (A_1, A_0)$, $B = (B_1, B_0)$
- $A = 2^{n/2} A_1 + A_0$
- $B = 2^{n/2} B_1 + B_0$
- $AB = (2^{n/2} A_1 + A_0)(2^{n/2} B_1 + B_0)$
- $T(n)$ = # of steps to multiply 2 n-bit numbers
- $T(\frac{n}{2})$ = # of steps to multiply two $\frac{n}{2}$ bit numbers
- $T(n) = 4T(n/2) + 4n$

$a = 4$, $b = 2$, $\log_2 4 = 2$, $f(n) = 4n$

$n^2 = O(n^{-\varepsilon})$ $<$ case 1

- $A \times B$

$\Theta(n^{\log_2 3})$

-- Merge Sort

- $T(n)$ = # of steps to do merge sort for list of size n

merge (mergesort $\left(A, \text{first}, \frac{\text{first}+\text{last}}{2}\right)$, $\left(A, \frac{\text{first}+\text{last}+1}{2}, \text{last}\right)$)

$T(n/2)$ $T(n/2)$

running time = $2T(n/2) + n$ case 2